

# Jasper Sands

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## RESEARCH INTERESTS

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Quantum error correction and decoding · quantum algorithms and query complexity · machine learning for quantum-device characterization and control.

## EDUCATION

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**Columbia University**, New York, NY Sep 2025 – May 2027

*Master of Science in Computer Science, Quantum Computing Track.* GPA 3.95/4.0.

**Thesis:** *Automation of Silicon Spin Qubit Measurements Using Machine Learning.* Advised by Dr. Nard Dumoulin Stuyck at Diraq and UNSW Sydney, with Dr. Henry Yuen at Columbia.

**Coursework:** Quantum Error Correction · Quantum Engineering · Quantum Simulation & Computing Lab · Quantum Optimization & Machine Learning · Neural Networks & Deep Learning · Machine Learning for Functional Genomics.

**Washington University in St. Louis**, St. Louis, MO Aug 2021 – May 2025

*Bachelor of Science in Computer Science and Economics; Minor in Political Science.* GPA 3.81/4.0.

**Coursework:** Machine Learning · Analysis of Algorithms · Algorithms for Biosequence Comparison.

## TECHNICAL REPORTS

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Kate Bonner and Jasper Sands. *Noise-Aware Ansatz Design for Digitized Adiabatic Quantum Computing.* 2025.

Mike Azure and Jasper Sands. *Practical LLM-Based Lossless Compression: An Empirical Study.* 2025.

John C. Papciak and Jasper Sands. *Multimodal Hyperbolic Embeddings for AMR Detection and Taxonomic Modeling in Metagenomic Contigs.* 2025.

## RESEARCH EXPERIENCE

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**Diraq and UNSW Sydney** · *MS Thesis Research* Sep 2026 – May 2027

- Automating silicon spin-qubit characterization under Dr. Nard Dumoulin Stuyck, targeting manual tuning and interpretation of charge measurements that bottleneck semiconductor quantum-hardware development.
- Building pipelines that acquire charge-measurement data from cryogenic probing infrastructure and classify charge transitions and device states with machine learning.
- Benchmarking automated classification against expert manual interpretation of the same device measurements.

**Cidon Systems Research Lab**, Columbia University · *Graduate Researcher* Sep 2025 – Jun 2026

- Built phishing and spam detection pipelines with Dr. Asaf Cidon and Barracuda Networks, combining language-model embeddings with production email telemetry and optimizing precision under a fixed false-positive budget.

**Desdr Open Insurance Toolkit**, Columbia University · *Graduate Researcher* Sep 2025 – Dec 2025

- Built climate-risk and actuarial features from sparse SMS survey data with Dr. Eugene Wu, modeling expected agricultural losses for micro-insurance program design.

**Complex Resilient Intelligent Systems Lab**, Columbia University · *Graduate Researcher* May 2025 – Dec 2025

- Reduced hallucination in generative protein modeling with Dr. Venkat Venkatasubramanian by grounding model outputs in ontology and molecular-graph data.

**Foster System Database**, Washington University in St. Louis · *Undergraduate Researcher* Sep 2024 – May 2025

- Fine-tuned LLaMA-3 under Dr. Ian Fillmore for question answering over 50k foster-care policy documents, with human-in-the-loop feedback for cross-state policy analysis.

## SELECTED PROJECTS

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**WhatTheDuck:** quantum amplitude estimation for Value at Risk · *CUDA-Q, C++* · MIT iQuHACK 2026

- Wrote GPU-accelerated state preparation, threshold oracle, and bisection search for Iterative Quantum Amplitude Estimation; measured query scaling matched the  $O(1/\epsilon)$  bound against an  $O(1/\epsilon^2)$  classical baseline.

**WASM QEC Simulator:** surface-code decoding in the browser · *Rust, WebAssembly*

- Built a browser-native symplectic stabilizer simulator for rotated and XZZX surface codes with real-time syndrome visualization and MWPM/Union-Find decoding; finite-size scaling estimated thresholds of 3.2% and 2.7% respectively.

**Q-Search:** multi-agent quantum algorithm research and verification engine

- Built a multi-agent quantum-algorithm research engine that proposes candidate algorithms, verifies circuits through simulation, benchmarks query and gate complexity against 1,200+ classical baselines, and indexes falsified approaches across hidden-subgroup, dihedral-coset, and code-equivalence problems.

**Noise-Aware Adiabatic Simulation:** transverse-field Ising model · *Qiskit* · IBM Heron r2

- Digitized adiabatic ground-state preparation on IBM Heron r2 using its parameterised RZZ( $\theta$ ) pulses rather than fixed two-qubit decompositions, with palindromic Trotter steps for second-order accuracy.
- Flattened the annealing schedule near the critical point to slow evolution as the spectral gap closes.

## TEACHING EXPERIENCE

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**Introduction to Quantum Computing**, Columbia University · *Teaching Assistant* Jan 2026 – May 2026

**Parallel and Concurrent Programming**, Washington University · *Teaching Assistant* Jan 2025 – May 2025

**Reverse Engineering and Malware Analysis**, Washington University · *Teaching Assistant* Aug 2024 – Dec 2024

## INDUSTRY EXPERIENCE

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**Smack Technologies** · *AI Intern* May 2026 – Aug 2026

- Built physics and sensor simulations for core AI pipelines, integrating knowledge-graph state representations to expand synthetic corpus generation.
- Benchmarked open-weight models across task performance, inference speed, and concurrency to increase generation throughput 64×, and validated simulator outputs against independent references within 0.55% error.

**Highnote** · *Cybersecurity Engineer Intern* May 2024 – Jul 2024

- Designed and deployed the company-wide SIEM for a card-issuing fintech, ingesting 1+ TB/day and supporting 100+ rule-based and ML anomaly detections.

**Mindtrip** · *Software Engineer Intern* May 2023 – Jul 2023

- Built internal admin and secure-query tooling with Retool, JavaScript, and Ruby on Rails, saving roughly 250 engineering hours/month.

## AWARDS AND COMPETITIONS

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- **Overall Winner & NVIDIA Ecosystem Award**, MIT iQuHACK 2026, State Street/Classiq Challenge.
- **Qualified**, NYU Abu Dhabi Quantum and AI Hackathon for Social Good 2027.
- **Audience Favorite**, Harmoniqs Quantum Design Hackathon 2026.
- **Runner-Up**, Qualcomm Snapdragon Multiverse Hackathon 2026.

## TECHNICAL SKILLS

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**Quantum software:** Qiskit, Qiskit Runtime, Cirq, PennyLane, CUDA-Q, Stim, PyMatching, QuTiP · statevector and noise-aware simulation, transpilation, backend execution, benchmarking.

**Quantum algorithms:** Hamiltonian simulation, quantum phase estimation, amplitude amplification and estimation, QSVT, block encoding, quantum walks, VQE, QAOA, Trotter decomposition.

**Quantum error correction:** Rotated and XZZX surface codes, stabilizer formalism, MWPM and Union-Find decoding, finite-size scaling, threshold estimation, noise modeling.

**Programming:** Python, C++, Rust, C, MATLAB, JavaScript/TypeScript, Go, Java.

**ML & systems:** PyTorch, TensorFlow, scikit-learn, Hugging Face, CUDA, NumPy, SciPy, Linux, Git, Docker, WebAssembly, AWS, GCP, LaTeX.

## LEADERSHIP AND SERVICE

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- **Graduate Lead**, Columbia Quantum Algorithms Reading Group, Sep 2025 – Jun 2026. Led weekly sessions covering all 33 chapters of Andrew Childs' quantum algorithms notes, paired with simulation checks.
- **Founding-semester member**, Columbia Autonomous Racing Club. Coordinate sponsor outreach and track testing.
- **Volunteer Tutor**, St. Louis Tutor Me Program. K-12 mathematics and literacy in underserved districts.